POWER CONSUMPTION ANALYSIS

Robot Power & Distribution System

Team ARDILLA

Elsa OUMA | Martina ESQUIROZ

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1. Introduction

This document presents the complete power consumption analysis for the ARDILLA robot. Its purpose is to verify that the power architecture — based on two 11.1 V LiPo batteries — is correctly dimensioned to supply all electronic, computational, and mechanical subsystems safely and reliably.

The analysis covers all components listed in the Bill of Materials (BOM V2, May 2026). For each component, the operating voltage, typical current draw, and power consumption are stated. The document concludes with a full power budget summary, battery runtime estimates, and a fuse safety verification.

2. Methodology

Power consumption is calculated using Ohm's Law:

$$P\ (W) = V\ (V) \times I\ (A)$$

Where P is power in Watts, V is the operating voltage, and I is the typical current drawn by the component. Values are taken from manufacturer datasheets or well-established community benchmarks where datasheets are not publicly available.

Three power rails are defined for this system:

- 12 V rail — directly from the LiPo battery pack via the XT60 connectors and 15A fuse. Supplies the drive motors and motor driver.
- 5 V rail — supplied by two step-down DC-DC converters (BOM #9). One converter powers the logic electronics (Horizon X3 Pi, RPLIDAR, Arduino Mega); the other is reserved for servo supply.
- 3.3 V rail — derived from the 5 V rail via the LD1117V33 LDO regulator on the Sensor/IO board. Supplies I2C sensors, ADC, and the ultrasonic module.

For runtime estimation, the 80% usable capacity rule is applied to the LiPo batteries, meaning only 80% of the total stored energy is used to protect battery chemistry and avoid deep discharge:

$$\text{Runtime (h)} = \text{Usable Energy (Wh)} \div \text{Total Power (W)}$$

3. Component Power Analysis

3.1 Power Source — LiPo Batteries (#4)

The robot is powered by two RS PRO LiPo batteries rated at 2.6 Ah and 11.1 V each, connected in parallel. This doubles the capacity while maintaining the nominal voltage.

Parameter	Value	Notes
Nominal voltage	11.1 V	3-cell LiPo (3 × 3.7 V)
Capacity per cell	2.6 Ah	RS PRO specification
Total capacity (parallel)	5.2 Ah	2 × 2.6 Ah
Total energy	57.7 Wh	5.2 Ah × 11.1 V
Usable energy (80%)	46.2 Wh	Safe discharge limit for LiPo

3.2 Drive Motors — RS PRO DC Gear Motor (#10, ×2)

Two RS PRO 12V DC gear motors (10 Ncm, 180 rpm, 5 mm shaft) provide traction. These are the dominant consumers in the power budget, driven through the DFRobot motor driver on the 12 V rail.

Parameter	Rail	Voltage (V)	Current (A)	Power (W)
Motor M1 — typical running	12 V	11.1	1.00	11.1
Motor M2 — typical running	12 V	11.1	1.00	11.1
Both motors — heavy load	12 V	11.1	2.00 each	44.4
Both motors — stall (peak)	12 V	11.1	4.00 each	~88.8

NOTE: Stall current should be avoided during normal operation. The 15A fuse protects against prolonged stall, but the motors themselves may overheat if held stalled for extended periods.

3.3 Motor Driver — DFRobot Dual-Channel 10A (#11)

The DFRobot DFR0601 dual-channel MOSFET-based motor driver provides up to 10A per channel, making it well-suited for these motors. It consumes a small amount of power itself through logic supply and switching losses.

Component	Rail	Voltage (V)	Current (A)	Power (W)	Notes
Driver logic (VCC)	5 V	5.0	0.05	0.25	Control circuitry
MOSFET switching loss	12 V	11.1	~0.12	~1.3	~6% of motor load

3.4 Computing — Horizon X3 Pi (#1)

The Horizon RDK X3 (Sunrise X3 Pi) is the robot brain. It runs Ubuntu 20.04 and handles high-level decision making, AI inference (5 TOPS), sensor fusion, and communication. The official power specification requires a 5V/3A supply, and typical consumption under a robot workload is approximately 4.5 W.

Operating mode	Rail	Voltage (V)	Current (A)	Power (W)
Idle	5 V	5.0	0.60	3.0
Typical robot workload	5 V	5.0	0.90	4.5
Peak (AI inference + I/O)	5 V	5.0	up to 1.50	up to 7.5

3.5 Microcontroller — Arduino Mega 2560 R3 (#3)

The Arduino Mega 2560 R3 handles low-level control tasks including PWM signals to the motor driver and reading encoder feedback. It operates from the 5V logic rail and consumes minimal power.

Component	Rail	Voltage (V)	Current (A)	Power (W)	Notes
Arduino Mega 2560 R3	5 V	5.0	0.15	0.75	Board + active I/O

3.6 LIDAR Scanner — RPLIDAR A1 (#12)

The RPLIDAR A1 is a 360-degree 2D laser scanner used for localisation and obstacle detection. It draws approximately 700 mA at 5V during continuous scanning operation, as documented in the Slamtec datasheet.

Component	Rail	Voltage (V)	Current (A)	Power (W)	Notes
RPLIDAR A1 (scanning)	5 V	5.0	0.70	3.50	Datasheet: ~700mA
RPLIDAR A1 (idle/motor off)	5 V	5.0	0.10	0.50	Motor stopped

3.7 Camera — HBV-313 USB Camera (#13)

The HBV-313 is a USB camera module from HBVCAM, used for vision processing. Based on published power data from HBVCAM USB modules, it consumes approximately 150–200 mW during active imaging.

Component	Rail	Voltage (V)	Current (A)	Power (W)	Notes
HBV-313 camera (active)	5 V	5.0	0.04	0.20	Via USB from X3 Pi

3.8 Servo Motor — JAMARA High-Torque Servo (#14, ×1)

One high-torque servo motor (4–6 V operating range) is used for mechanism actuation. The servo supply is provided by the second DC-DC converter (BOM #9), which is kept separate from the logic supply to prevent motor noise from resetting the X3 Pi.

Operating state	Rail	Voltage (V)	Current (A)	Power (W)	Notes
Servo — active / moving	5 V	5.0	0.50	2.50	Under moderate load
Servo — holding position	5 V	5.0	0.10	0.50	Static hold

3.9 DC-DC Converters (#9, ×2)

Two step-down (buck) DC-DC converters step the 11.1V battery voltage down to 5V. One supplies the logic and sensors, the other supplies the servo. Assuming 90% conversion efficiency, approximately 10% of the output power is lost as heat in each converter.

Converter	Input	Output	Load (W)	Loss at 90% eff. (W)
#9a — Logic supply (X3 Pi, RPLIDAR, Arduino, camera)	11.1 V	5 V	~8.9	~0.99
#9b — Servo supply (JAMARA servo)	11.1 V	5 V	~2.5	~0.28

DESIGN NOTE: Using two separate converters for logic and servo power is good practice. Servo motors generate voltage spikes when reversing direction, which can interfere with sensitive logic. Separate rails provide isolation.

3.10 Sensor & I/O Board Components

The Sensor/IO board (from the KiCad schematic) contains several low-power ICs running on the 3.3V rail. These collectively represent a negligible fraction of the total power budget.

Component	Rail	Voltage (V)	Current (mA)	Power (W)	Function
LD1117V33 LDO (U4) — heat loss	5V→3.3V	—	—	0.85	(5-3.3V)×0.5A wasted
TCA9548A I2C multiplexer (U2)	3.3 V	3.3	5	0.016	I2C bus routing
ADS1115 16-bit ADC (U3)	3.3 V	3.3	0.15	0.0005	Analog readings
INA219 current monitor	5 V	5.0	1	0.005	12V rail monitoring
Ultrasonic sensor (J2)	3.3 V	3.3	15	0.050	Distance sensing
Limit switches ×2 (#15)	3.3 V	3.3	~1	0.003	End-stop detection
SD Card (#2)	5 V	3.3	50	0.15	Via X3 Pi internally

3.11 Passive Components (Switches, Fuse, Connectors)

The main ON/OFF switch (#5), emergency stop button (#6), ATC fuse and holder (#7), and XT60 connectors (#8) are all passive components that carry current but consume no power themselves in normal operation.

FUSE NOTE: The 15A slow-blow ATC fuse (#7) is correctly rated. At typical operation (~4A from the battery), it provides a 3.75× safety margin. It will only blow in the event of a short circuit or sustained overcurrent above 15A.

4. Complete Power Budget Summary

The following table consolidates the power consumption of all BOM components across the three power rails. Values are for typical operating conditions with both motors running at moderate speed.

#	Component	Rail	V (V)	I (A)	P typical (W)	P peak (W)
1	Horizon X3 Pi	5 V	5.0	0.90	4.50	7.50
2	SanDisk SD Card 64GB	5 V	3.3	0.05	0.15	0.20
3	Arduino Mega 2560 R3	5 V	5.0	0.15	0.75	0.80
4	LiPo Batteries ×2	—	11.1	—	Source	Source
5	Main ON/OFF Switch	12 V	—	~0	~0	~0
6	Emergency Stop Button	12 V	—	~0	~0	~0
7	Fuse + Holder (15A)	12 V	—	~0	~0	~0
8	XT60 Connectors	12 V	—	~0	~0	~0
9a	DC-DC Converter (logic) loss	12 V	11.1	~0.09	0.99	1.30
9b	DC-DC Converter (servo) loss	12 V	11.1	~0.03	0.28	0.40
10	RS PRO DC Gear Motors ×2	12 V	11.1	1.00 ea.	22.20	88.80
11	DFRobot Motor Driver (10A)	5V+12V	—	—	1.55	2.00
12	RPLIDAR A1	5 V	5.0	0.70	3.50	3.50
13	HBV-313 USB Camera	5 V	5.0	0.04	0.20	0.25
14	JAMARA Servo Motor ×1	5 V	5.0	0.50	2.50	3.00
15	Limit Switches ×2	3.3 V	3.3	~0.001	~0	~0
—	Sensor/IO board ICs	3.3 V	3.3	~0.02	~0.15	0.20
	TOTAL	All	—	—	**~36.8 W**	**~108 W**

4.1 Power by Rail

Rail	Typical Power (W)	Peak Power (W)	Supplied by
12 V (motors + driver)	~24.0	~90.8	LiPo batteries directly
5 V (logic + sensors + servo)	~12.6	~16.2	2× DC-DC step-down converters
3.3 V (I2C sensors, ADC)	~0.15	~0.20	LD1117V33 on Sensor/IO board
TOTAL	~36.8 W	~108 W	

5. Battery Runtime Estimation

Using the usable energy of 46.2 Wh (80% of 57.7 Wh total), the estimated battery runtime under three representative operational scenarios is:

Scenario	Total Power (W)	Estimated Runtime	Description
Electronics only (no motors)	~6.0 W	~7.7 hours	X3 Pi + sensors idle, motors off
Normal operation	~36.8 W	~1.25 hours	Both motors at moderate speed, all sensors active
Heavy load	~55.0 W	~0.84 hours	Motors under high load, all systems active
Peak / stall (brief)	~108 W	~0.43 hours	Should not be sustained — fuse protection active

For a competitive robotics context, a runtime of approximately 1.25 hours under normal operating conditions is adequate for extended testing and multiple runs. The two batteries in parallel provide both the capacity and current capability required.

6. Protection and Safety Verification

6.1 Fuse Check — ATC 15A Slow-Blow (#7)

Scenario	Battery current draw (A)	Status
Electronics only (no motors)	~0.54 A	Safe (3% of fuse rating)
Normal driving (both motors)	~3.8 A	Safe (25% of fuse rating)
Both motors heavy load	~5.0 A	Safe (33% of fuse rating)
Both motors stalled (peak)	~9.7 A	Safe (65% of rating) — avoid prolonged stall
Short circuit	> 15 A	Fuse blows — circuit protected

CONCLUSION: The 15A slow-blow fuse is correctly specified. Under all normal operating scenarios, the current draw is well within the fuse rating, providing appropriate protection without nuisance tripping.

6.2 DC-DC Converter Sizing

The two DC-DC converters must be sized for the maximum 5V load they will supply:

- Logic converter (#9a): Max load = X3 Pi (1.5A peak) + Arduino (0.15A) + RPLIDAR (0.7A) + Camera (0.04A) = ~2.4A. A converter rated at 3A minimum is required.
- Servo converter (#9b): Max load = 1 servo (0.5A typical, up to 1A peak) = 1A. A converter rated at 2A minimum is sufficient.

7. Conclusions

The power architecture of the ARDILLA robot is correctly dimensioned for the components defined in BOM V2 (May 2026). The key conclusions are:

- Total typical power consumption is approximately 36.8 W, dominated by the two 12V drive motors (60% of budget).
- The two LiPo batteries in parallel (57.7 Wh total, 46.2 Wh usable) provide an estimated runtime of approximately 1.25 hours under normal conditions.
- The 15A slow-blow fuse provides adequate circuit protection across all operating scenarios.
- The two DC-DC converters correctly replace the LM7805 LDO and are necessary to supply the X3 Pi and servo with sufficient current.
- Keeping the servo on a separate 5V rail from the logic electronics is good engineering practice and eliminates potential noise-induced resets.
- The Sensor/IO board (3.3V rail) consumes a negligible fraction (<0.5%) of the total power budget.